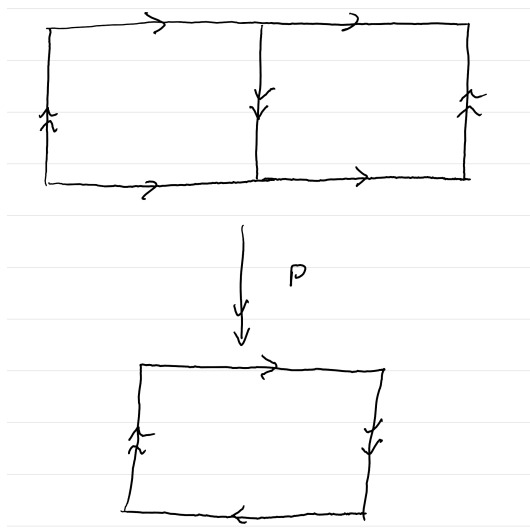
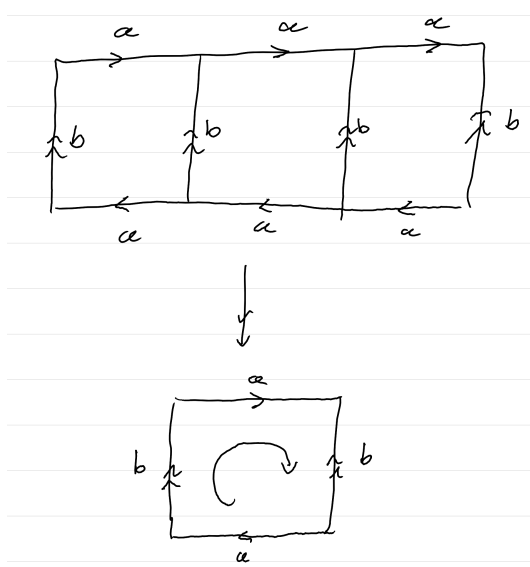
Revised Problems

Problem (Homework 17, Problem 1)**:** Give two examples of a nontrivial covering space of the Klein bottle, one normal and one non-normal.

Solution: As exhibited below, we can view the torus as a two-sheeted cover of the Klein bottle. Since the number of sheets is equal to the index of $p_*\left(\pi_1\left(\widetilde{X}, \widetilde{x}_0\right)\right)$ in $\pi_1(X, x_0)$, and the covering map is injective, it follows that this is a normal covering since index 2 subgroups are normal.



Similarly, to find a non-normal covering, we consider the following three-sheeted covering space. This covering lifts the loop given by a to three copies of itself, while it lifts b to one copy of itself. In particular, this means that this cover corresponds to the subgroup of the Klein bottle group $\left\langle a, b | abab^{-1}\right\rangle$ given by $H = \left\langle a^3, b\right\rangle$. Yet, since $b \in H$, but $aba^{-1} = a(aba)a^{-1} = a^2b$, it follows that H is not normal, so this is a non-normal covering space of the Klein bottle.



Current Problems

Problem (Problem 1)**:** Let $\left[v_0,\dots,v_q\right]$ be a q-simplex in $\mathbb{R}^m$ for $m \geq q$. Prove that if d is the diameter of a simplex in the barycentric subdivision of $\left[v_0,\dots,v_q\right]$, then $d \leq \frac{q}{q+1}\operatorname{diam}\left(\left[v_0,\dots,v_q\right]\right)$.

Solution: We claim that

$$\operatorname{diam}\left(\left[v_0,\dots,v_q\right]\right) = \max\left\{\left|v_i - v_j\right| : 1 \leq i < j \leq q\right\}.$$

This follows from the fact that if u and w are any two elements of Δ^q, and w has convex combination expression

$$w = \sum_{i=0}^{q} s_i v_i,$$

then

$$\begin{aligned}
|u - w| &= \left|u - \sum_{i=0}^{q} s_i v_i\right| \\
&= \left|\sum_{i=0}^{q} s_i (u - v_i)\right| \\
&\leq \sum_{i=0}^{q} s_i |u - v_i| \\
&\leq \sum_{i=0}^{q} s_i \max_{i=0}^{q} |u - v_i| \\
&= \max_{i=0}^{q} |u - v_i|.
\end{aligned}$$

In particular, this means that if $u \in \Delta^q$, then $|u - v_i| \leq \max\left\{\left|v_i - v_j\right| : 1 \leq i < j \leq q\right\}$. Thus, this is in fact our desired diameter.

Observe then that, if we set

$$b = \frac{1}{q+1}\sum_{i=0}^{q} v_i,$$

then for any v_j, we have

$$\begin{aligned}
\left|b - v_j\right| &= \left|\frac{1}{q+1}\sum_{i=0}^{q} v_i - v_j\right| \\
&= \left|\frac{1}{q+1}\sum_{i=0}^{q}\left(v_i - v_j\right)\right| \\
&\leq \frac{1}{q+1}\sum_{i=0}^{q}\left|v_i - v_j\right| \\
&= \frac{1}{q+1}\sum_{i\neq j}\left|v_i - v_j\right| \\
&\leq \frac{1}{q+1}\sum_{i\neq j}\max\left\{\left|v_i - v_j\right| : 1 \leq i < j \leq q\right\} \\
&= \frac{q}{q+1}\operatorname{diam}\left(\left[v_0,\dots,v_q\right]\right).
\end{aligned}$$